WRB: Introduction

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World Reference Base for Soil Resources (WRB)

Contents:

1. History
2. How the system works
3. How to classify soils in the Soil Judging Contest 2026
4. Questions (and answers)

World Reference Base for Soil Resources (WRB)

1. History

International soil classification system

World Reference Base for Soil Resources (WRB)

2 roots:

- Legend of the Soil Map of the World (1 : 5 M)
  (FAO: Food and Agriculture Organization of the United Nations)
- IRB: International Reference Base for Soil Classification
  (ISSS: International Society of Soil Science)

Legend of the Soil Map of the World

1974: first edition
26 soil groups
FAO-UNESCO (1971-1981): Soil Map of the World, 1 : 5 M, Vol. 1-10, Paris.

1988: second edition: Revised Legend
28 soil groups

map:
FAO/IIASA/ISRIC/ISSCAS/JRC (2012): Harmonized World Soil Database (version 1.2).
FAO, Rome, Italy, and IIASA, Laxenburg, Austria.
https://webarchive.iiasa.ac.at/Research/LUC/External-World-soil-database/HTML/
(IIASA: International Institute for Advanced System Analysis)

legend of a soil map, but suitable also for soil classification
grouping of soils predominantly according to their land use potential

IRB

1980: initiated by soil scientists who were unsatisfied with the FAO Legend
aim: grouping of soils more according to their natural genesis

1982: IRB established as ISSS Working Group:
16 soil groups

1990: revision of the IRB:
20 soil groups

The road to the WRB

FAO 1974:
26 soil groups

IRB 1982:
16 soil groups

FAO 1988:
28 soil groups

IRB 1990:
20 soil groups

1992: decision to create the WRB system:
based on the 28 FAO soil groups
using more genetic elements in the definitions

WRB

1998: first edition:
30 Reference Soil Groups
official classification system of ISSS and FAO

2006: second edition:
32 Reference Soil Groups

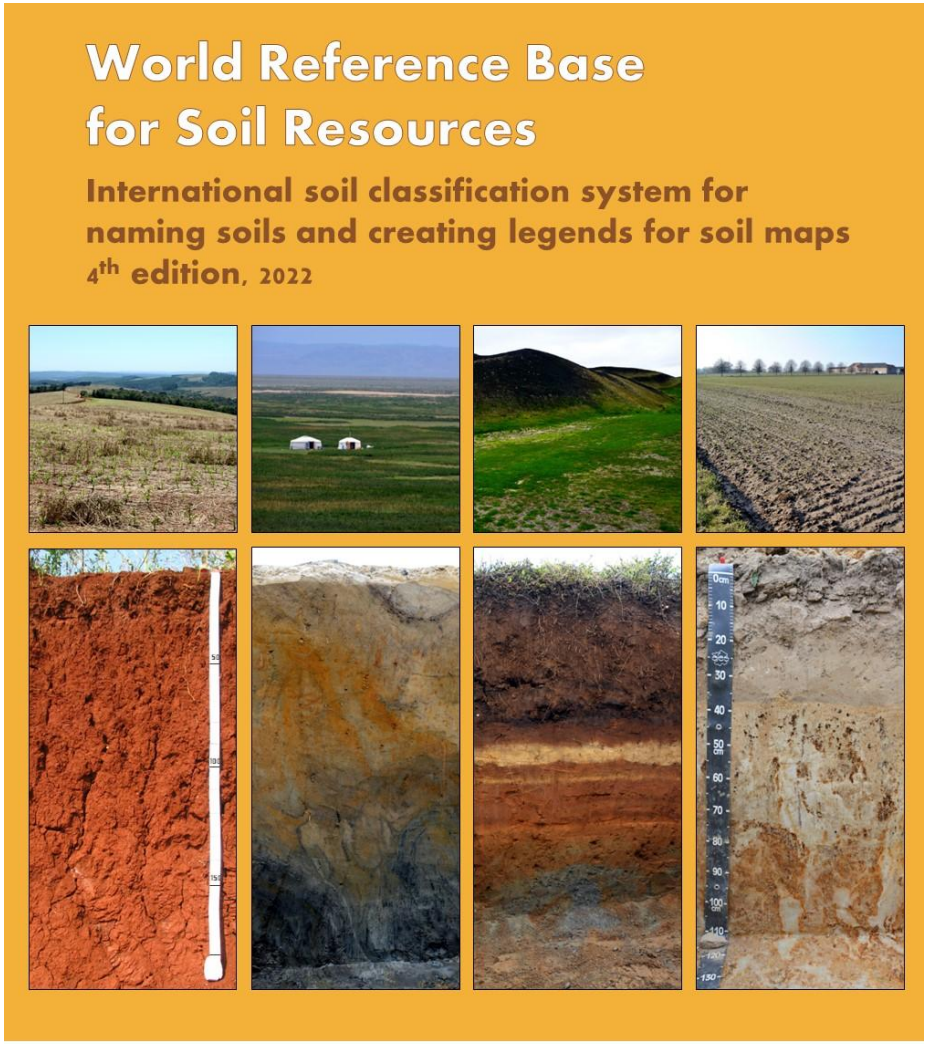
2010: Guidelines for constructing small-scale map legends using the WRB

2014 (Update 2015): third edition
32 Reference Soil Groups

2022: fourth edition
32 Reference Soil Groups

(2002: ISSS renamed into IUSS: International Union of Soil Sciences)

WRB, fourth edition, 2022



IUSS Working Group WRB (2022):
World Reference Base for Soil Resources
4th edition, chaired by Peter Schad and Stephan Mantel
International Union of Soil Sciences, Vienna.

WRB homepage: https://wrb.isric.org

WRB manual:
https://wrb.isric.org/files/WRB_fourth_edition_2022-12-18_errata_correction_2024-09-24.pdf

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article explaining WRB:
Peter Schad: World Reference Base for Soil Resources - Its fourth edition and its history.
Journal of Plant Nutrition and Soil Science (2023)
https://onlinelibrary.wiley.com/doi/10.1002/jpln.202200417
(open access)

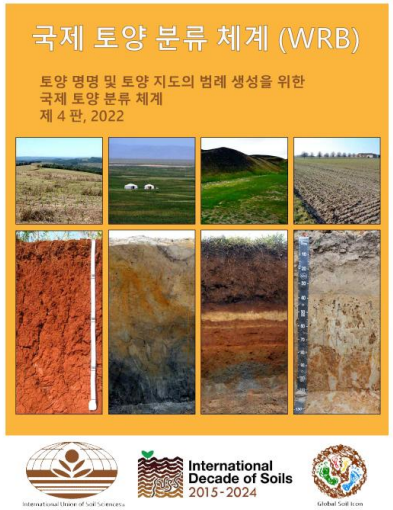
International Union of Soil Sciences®

International Decade of Soils 2015-2024

Global Soil Icon

The 4th edition of the WRB: translations

Italian



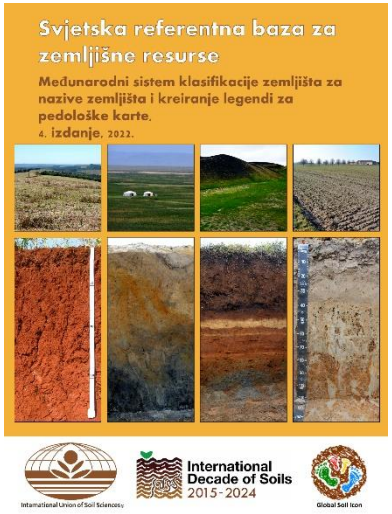
Korean

French

Spanish

Catalan

Portuguese



Serbo-Croatian

in preparation:
Russian
Bahasa Indonesia
Polish
Georgian
Chinese

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2. How the system works

WRB: worldwide common language of soil classification

soil scientists all over the world classify their soils according to 2 systems:
- according to their national system (if existing)
- according to WRB

impossible:
the simple translation of soil groups (soil types, soil units etc.)
between a national classification system (currently existing around 50) and the WRB:
- different architecture
- different threshold values for definitions
→ classification has to be based on the primary data

The architecture of the WRB

first classification level:
Reference Soil Groups (RSGs)

second classification level:
names of qualifiers
added to the name of the RSG

The first classification level

32 Reference Soil Groups (RSGs)
identified using a key (Chapter 4 of the WRB manual):

Histosol	Chernozem
Anthrosol	Kastanozem
Technosol	Phaeozem
Cryosol	Umbrisol
Leptosol	Durisol
Solonetz	Gypsisol
Vertisol	Calcisol
Solonchak	Retisol
Gleysol	Acrisol
Andosol	Lixisol
Podzol	Alisol
Plinthosol	Luvisol
Planosol	Cambisol
Stagnosol	Fluvisol
Nitisol	Arenosol
Ferralsol	Regosol

The second classification level

202 qualifiers: defined in a specific chapter (Chapter 5 of the WRB manual)
some can be combined with many RSGs, others with a few ones or with just one

for every RSG:
individual list of available qualifiers (Chapter 4 of the WRB manual, together with the key)
subdivided into principal qualifiers and supplementary qualifiers
(some qualifiers: principal for some RSGs, supplementary for others)

per RSG: 40 to 79 qualifiers
(Nitisols: 40; Gleysols: 79)
many qualifiers: mutually exclusive; most soils: < 10 qualifiers

Qualifiers: indicators of fertility and management

soil fertility:
texture: Arenic, Siltic, Clayic, Loamic
coarse fragments: Coarsic, Skeletic
thin soils: Leptic
dominance of exchangeable base cations or of exchangeable Al ions: Eutric, Dystric
ploughed: Aric

threats:
contamination: Toxic
compaction: Densic
accumulation: Solimovic, Novic

How are the RSGs and the qualifiers defined?

diagnostics: diagnostic horizons, diagnostic properties and diagnostic materials:
+- long definitions
defined in a specific chapter (Chapter 3 of the WRB manual)
have a name

questions in the key to the RSGs (first classification level; Chapter 4 of the WRB manual)
and definitions of the qualifiers (second classification level; Chapter 5 of the WRB manual):

- presence or absence (at a certain depth)
  of certain (predefined) diagnostic horizons, properties or materials
- single characteristics which can be described with a short wording
  (not predefined; e.g. a minimum clay content or a minimum base saturation)

The diagnostics of the WRB

diagnostic materials:
materials that significantly influence soil-forming processes; they may
- stem from the parent material or
- be the result of soil-forming processes

diagnostic properties:
- are typical results of soil-forming processes or
- reflect special conditions of soil formation

diagnostic horizons:
are typical results of soil-forming processes, and have a minimum thickness
→ recognizable as horizontal layer

attention: not all the horizons in a soil profile are diagnostic

Overlapping and coinciding diagnostics

A layer may fulfil the criteria of more than one diagnostic horizon, property or material, which are then regarded as overlapping or coinciding

example:
0 - 30 cm: mollic horizon
20 -50 cm: calcic horizon

example:
0 - 100 cm: calcaric material
20 - 80 cm: protocalcic properties
40 - 60 cm: calcic horizon
60 - 80 cm: petrocalcic horizon

Classifying soils

for every RSG:
  individual list of available qualifiers:
  subdivided into principal qualifiers and supplementary qualifiers
check the lists from top to down
  add **all** applying qualifiers

principal qualifiers
ranked, listed according to their importance
added before the name of the RSG
from right to left
(the first qualifier in the list is placed closest to the name of the RSG)

supplementary qualifiers
first: qualifiers related to texture (if more than one applies, from top to down the profile)
then: all other qualifiers in alphabetical order (not ranked)
added after the name of the RSG, in brackets and separated from each other by commas
from left to right
(the first qualifier in the list is placed closest to the name of the RSG)

Qualifiers that are not listed

if your soil has qualifiers that are not in the list:
you may add them as the last supplementary qualifiers

common for polygenetic soils and for buried soils

The WRB Manual

1. Background and basics
2. The rules for naming soils and creating legends for soil maps
3. Diagnostic horizons, properties and materials
4. Key to the Reference Soil Groups with lists of principal and supplementary qualifiers
5. Definitions of qualifiers
6. Codes for the Reference Soil Groups, qualifiers and specifiers
7. References

8. Annex 1: Field Guide (replacing the FAO Guidelines for Soil Description, 2006)
9. Annex 2: Summary of analytical procedures for soil characterization
10. Annex 3: Horizon and layer designations (replacing the FAO Guidelines for Soil Description, 2006)
11. Annex 4: Soil description sheet
12. Annex 5: Guidance on database set-up
13. Annex 6: Colour symbols for RSG maps

pdf file on the WRB homepage

excel file on the WRB homepage

Classifying soils

parent material: glacial till

stagnic properties
(weakly expressed)

argic
horizon

parent material: dolomitic limestone

dolomitic material
(fine earth)

continuous rock
(limiting layer)

other characteristics:

- high CEC_{pot} per kg clay
- exchangeable base cations > exchangeable Al

argic horizon, high CEC_{pot} / kg clay, dominance of base cations, no other prominent characteristics → Luvisol

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Classifying soils: example Luvisol

list of available qualifiers:

principal qualifiers

Abruptic
Fragic
Petrocalcic
Leptic
Hydragric/
Anthraquic/
Irragric/ Pretic/
Terric
Gleyic
Stagnic
Vertic
Rhodic/
Chromic

Nudiargic
Lamellic
Albic
Ferric
Gypsic
Calcic
Yermic/ Takyric
Skeletic
Dolomitic/
Calcaric
Haplic

supplementary qualifiers

Arenic/ Clayic/
Loamic/ Siltic
Andic
Aric
Neocambic
/Neobrunic
Columnic
Cutanic
Densic
Differentic
Epidystric/
Hypereutric
Epic/ Endic
Escalic
Fluvic
Fractic
Gelic

Humic/ Ochric
Magnesic
Nechic
Nitic
Novic
Oxyaquic
Profondic
Pyric
Raptic
Sodic
Solimovic
Technic/ Kalaic
Toxic
Transportic
Turbic
Vitric

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Classifying soils: example Luvisol

list of available qualifiers (applying qualifiers in green):

principal qualifiers

Abruptic
Fragic
Petrocalcic
Leptic
Hydragric/
Anthraquic/
Irragric/ Pretic/
Terric
Gleyic
Stagnic
Vertic
Rhodic/
Chromic
Nudiargic
Lamellic
Albic
Ferric
Gypsic
Calcic
Yermic/ Takyric
Skeletic
Dolomitic/
Calcaric
Haplic

supplementary qualifiers

Arenic/ Clayic/
Loamic/ Siltic
Andic
Aric
Neocambic
/Neobrunic
Columnic
Cutanic
Densic
Differentic
Epidystric/
Hypereutric
Epic/ Endic
Escalic
Fluvic
Fractic
Gelic
Humic/ Ochric
Magnesic
Nechic
Nitic
Novic
Oxyaquic
Profondic
Pyric
Raptic
Sodic
Solimovic
Technic/ Kalaic
Toxic
Transportic
Turbic
Vitric

→ **Dolomitic Stagnic Leptic Luvisol (Clayic, Cutanic, Differentic, Epic, Humic, Raptic)**

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Classifying soils: example Luvisol: texture qualifiers

texture may change with soil depth

a texture qualifier applies
if the respective texture is present:

- in one or more layers with a combined thickness of ≥ 30 cm, occurring within 100 cm of the mineral soil surface, or
- in the major part between the mineral soil surface and a limiting layer starting > 10 and < 60 cm from the mineral soil surface

Loamic

Clayic

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Qualifiers: the redundance rule

redundant qualifiers are not added

examples:
- if Calcaric applies, do not add Eutric
- if Thaptomollic applies, do not add Panpaic

General definitions (see also Handbook of the Soil Contest)

Litter layer (WRB) or a **layer of freshly fallen plant litter** (ST):
- loose
- contains > 90% (by volume, related to the fine earth plus all dead plant residues) recognizable dead plant tissues (e.g., undecomposed leaves)

Soil surface (0 cm): the surface of the soil after removing, if present, the litter layer
Mineral soil surface: the upper limit of the uppermost layer consisting of mineral material (WRB) or mineral soil material (ST)

General definitions (specific for WRB)

If a criterion is worded as a conditional clause (if...) and the condition (**if-clause**) is not true, the criterion is ignored.

Numerical values obtained in the field or in the laboratory have to be taken as such and **must not be rounded** when compared with the threshold values in the diagnostic criteria.

The diagnostic criteria must be fulfilled **throughout the specified depth range**, unless stated otherwise. If a diagnostic horizon consists of several subhorizons, the diagnostic criteria (except thickness) must be fulfilled in every subhorizon separately (averages are not calculated), unless stated otherwise.

The term **limiting layer** used in definitions comprises *continuous rock*, *technic hard material, petrocalcic*, *petroduric*, *petrogypsic* and *petroplinthic horizons* and other cemented layers.

Subqualifiers 1

Endo	+	Skeletic	=	Endoskeletic
Specifier	+	*Qualifier*	=	*Subqualifier*

Types of subqualifiers:

1. Subqualifiers with a given definition (in Chapter 5)
2. Subqualifiers constructed by the user

2.1. Subqualifiers related to depth requirements: Epi-, Endo-, Amphi-, Ano-, Kato-, Poly-, Panto-, Bathy-

2.2. Subqualifiers related to other requirements

Supplementary qualifiers:
alphabetical sequence according to the qualifier, not the specifier

If the specifiers Proto- (weakly expressed), Bathy- (great depth) or Thapto- (buried) occur with principal qualifiers, the subqualifiers must shift to the supplementary qualifiers.

Subqualifiers 2

Subqualifiers related to depth requirements

Chapter 2.3

The rules for constructing a depth-related subqualifier depend on what the qualifier refers to:

1. qualifiers referring to a characteristic that occurs at a **specific point of depth** (e.g. Raptic): Epi-, Endo-, Amphi-
2. qualifiers referring to a **layer** (e.g. Calcic, Arenic, Fluvic): Epi-, Endo-, Amphi-, Ano-, Kato-, Poly-, Panto-
3. qualifiers referring to the **major part of a certain depth range or to half or more of a certain depth range** (Dystric and Eutric, only): Epi-, Endo-
4. qualifiers referring to a **specified depth range throughout** (Relocatic, only): Epi-
5. qualifiers referring to a **percentage** (e.g. Skeletic): Epi-, Endo-

Chapter 5

If a subqualifier related to depth requirements can be constructed by the user, **the figure indicates, which rule applies**: (1), (2), (3), (4), (5). If no figure is indicated, these subqualifiers cannot be constructed.

Examples:

Raptic (rp): having a *lithic discontinuity* at some depth ≤ 100 cm from the mineral soil surface, that is not related to *aeolic*, *fluvic, solimovic* or *tephric material* (1).

Calcic (cc): having a *calcic horizon* starting ≤ 100 cm from the mineral soil surface (2).

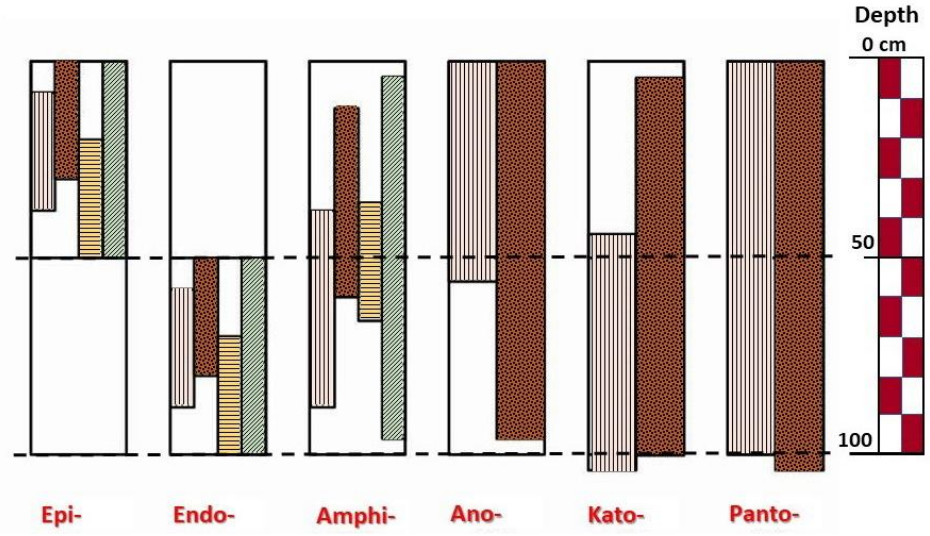
Cutanic (ct): having an *argic* or *natric horizon* that meets diagnostic criterion 2.b of the respective horizon.

Subqualifiers 3

Chapter 2.3.1

Subqualifiers related to depth requirements

Figure: Subqualifiers related to depth requirements and referring to a layer.



For every qualifier with depth requirements, the definition (Chapter 5) specifies whether the depth requirement refers **to the soil surface or to the mineral soil surface**.

Subqualifiers related to depth requirements are only used if the relevant soil characteristics are **reported until ≥ 100 cm of the (mineral) soil surface or to a limiting layer**, whichever is shallower, **or if the non-reported depth range does not influence the correctness of the qualifier**.

Poly-:

a. **diagnostic horizons**: a diagnostic horizon is present two or more times at the depth required by the qualifier definition, interrupted by layers that do not fulfil the criteria of the respective diagnostic horizon;
b. **other layers**: two or more layers within 100 cm of the (mineral) soil surface fulfil the criteria of the qualifier, interrupted by layers that do not fulfil the criteria of the qualifier;
   and the thickness criterion is fulfilled by the sum of the thicknesses of the layers; it may or may not be fulfilled by the single layers.

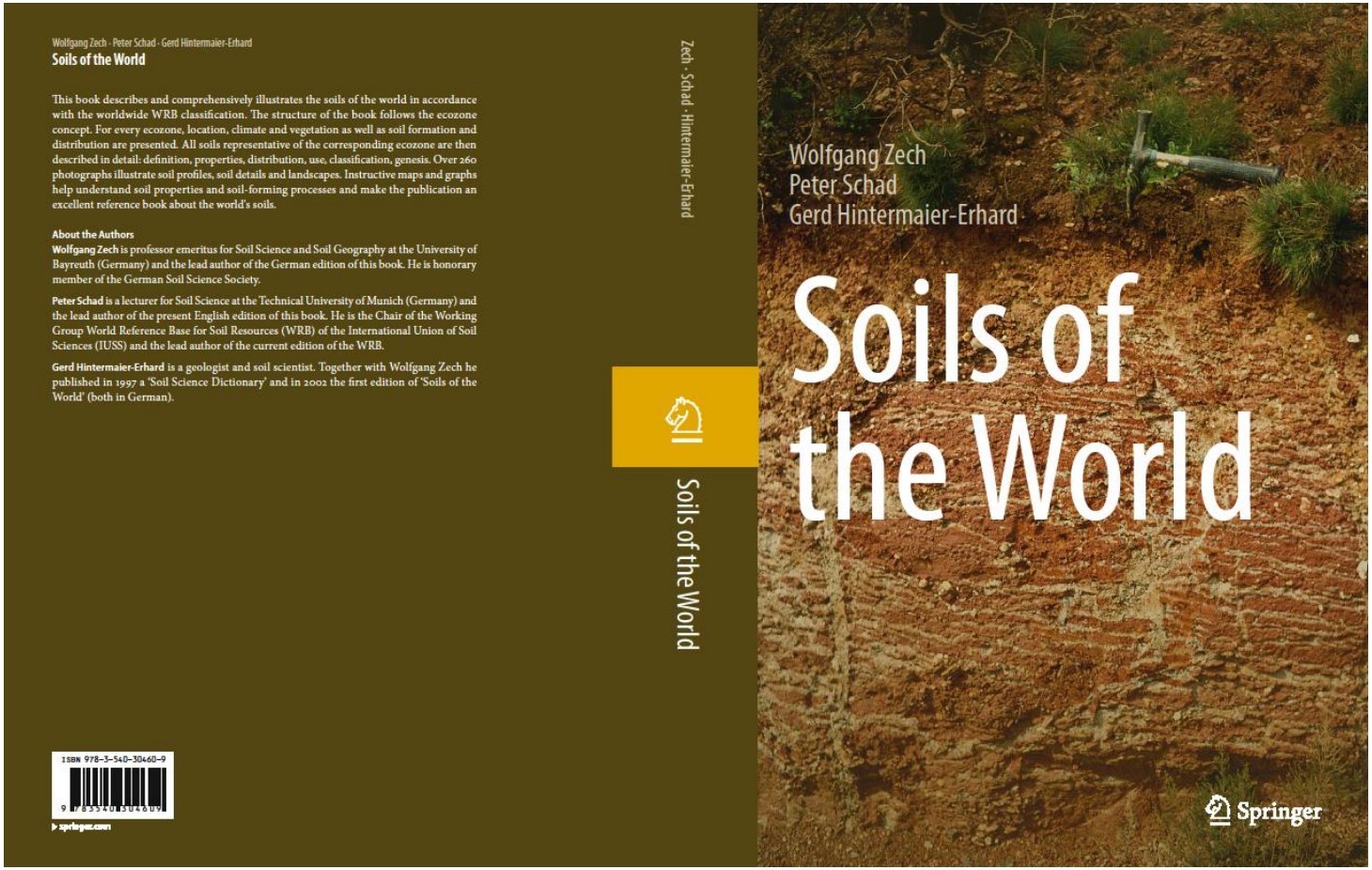
Bathy-: (shortened explanation)

- If the criteria of a qualifier are only fulfilled if layers at a depth of > 100 cm from the (mineral) soil surface are taken into account, the Bathy- specifier can be used to construct a subqualifier.
- The Bathy- subqualifier extends to a greater depth than indicated for the qualifier.

Literature

Soils according to WRB 2015:

Zech W, Schad P, Hintermaier-Erhard G (2022):
Soils of the World. Springer, Heidelberg.



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Literature

Photos according to WRB 2022:

Switoniak M et al. (2022):
Illustrated Handbook of WRB Soil Classification
https://www.researchgate.net/publication/368839948_Illustrated_Handbook_of_WRB_Soil_Classification

3. How to classify soils in the Soil Judging Contest 2026

Scoresheet: template, page 2

5. EVALUATION **SCORE:**

Suitability class	Wheat production	Paddy rice production	Home Gardens	Buildings without basement
Optimal				
Suitable				
Unsuitable				

6.1 CLASSIFICATION (WRB) **SCORE:**

Diagnostic horizons (10)	Diagnostic properties (5)	Diagnostic materials (5)	Reference Soil Group (15)	Qualifiers (5)

SCORE SUMMARY

Name examiner		
1.		
2.		
3.1		
3.2		
4.		
5.		
6.1		
6.2		
Total		

6.2 CLASSIFICATION (ST) **SCORE:**

Epipedon (10)	Subsurface horizons (5)	Diagnostic characteristics (5)	Order (10)	Suborder (5)	Great Group (5)	Subgroup (5)
Mollic						
Ochric						
Umbric						
None						

Final score.....

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Scoresheet: European Soil Judging Contest, Spain, 2025

V. HORIZON DESIGNATION AND SOIL CLASSIFICATION

V.1 WORLD REFERENCE BASE FOR SOIL RESOURCES (2022)

Diagnostic horizons (10)		Diagnostic properties (5)		Diagnostic materials (5)		Reference Soil Group (10)		Qualifiers (5)	
	Argic		Abrupt textural difference		Calcaric material		Arenosol		Calcaric
	Calcic		Continuous rock		Fluvic material		Calcisol		Calcic
	Cambic		Gleyic properties		Solimovic material		Cambisol		Cambic
	Mollic		Lithic discontinuity		None		Fluvisol		Chromic
	Panpaic		Protocalcic properties				Gleysol		Fluvic
	Petrocalcic		Reducing conditions				Kastanozem		Haplic
	Protovertic		Stagnic properties				Luvisol		Luvic
	Vertic		None				Phaeozem		Novic
	None						Regosol		Panpaic
							Stagnosol		Oxygleyic
							Vertisol		Petric
									Petrocalcic
									Reductigleyic
									Skeletic
									Stagnic
									Thaptomollic
									Transportic

The scoresheets distributed in Nanjing will be filled in, similarly.

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Classification according to WRB: first step: decide on diagnostics

Soil Judging website: Simplified versions of some WRB diagnostics:
https://raw.githubusercontent.com/soiljudgingiuss/soiljudgingiuss.github.io/main/sj_documents/IUSS_templates/WRB_simplified_versions_of_the_diagnostics_present_in_the_Nanjing_region.pdf

Only those diagnostics are provided that can be expected in the Nanjing contest region.

Simplified versions are always wrong:
They only consider the most widespread cases and ignore the rare ones.

If the simplified version and the official WRB Manual give different answers:
In this contest, both will be considered to be correct.

Classification according to WRB: first step: decide on diagnostics

Soil Judging website: Simplified versions of some WRB diagnostics:
https://raw.githubusercontent.com/soiljudgingiuss/soiljudgingiuss.github.io/main/sj_documents/IUSS_templates/WRB_simplified_versions_of_the_diagnostics_present_in_the_Nanjing_region.pdf

Table of Contents

Diagnostic horizons 2
1 Anthraquic horizon 2
2 Argic horizon 2
3 Cambic horizon 3
4 Hydragric horizon 3
5 Mollic horizon 4
6 Panpaic horizon 4
7 Protovertic horizon 5
8 Umbric horizon 5
9 Vertic horizon 6
Diagnostic properties 6
10 Abrupt textural difference 6
11 Continuous rock 7
12 Gleyic properties 7
13 Lithic discontinuity 7
14 Protocalcic properties 8
15 Reducing conditions 8
16 Shrink-swell cracks 8
17 Stagnic properties 9
Diagnostic materials 9
18 Calcaric material 9
19 Claric material 10
20 Solimovic material 10

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Classification according to WRB: second step: decide on the RSG

Use the WRB Manual:

https://wrb.isric.org/files/WRB_fourth_edition_2022-12-18_errata_correction_2024-09-24.pdf

4 Key to the Reference Soil Groups with lists of principal and supplementary qualifiers

Before using the key, please read the 'Rules for naming soils' (Chapter 2).

Key to the Reference Soil Groups	Principal qualifiers	Supplementary qualifiers
Soils having one or more of the following: 1. *organic material* starting ≤ 40 cm from the soil surface and having within 100 cm of the soil surface a combined thickness of: a. ≥ 40 cm if < 75% (by volume, related to the fine earth plus all dead plant residues) consists of moss fibres; *or* b. ≥ 60 cm; *or* 2. *organic material* starting at the soil surface, having a thickness of ≥ 10 cm and directly overlying ice, *continuous rock* or *technic hard material*; *or* 3. a layer of coarse fragments that, together with overlying *organic material*, if present, starts at the soil surface and has a thickness of a. ≥ 10 cm if overlying *continuous rock* or *technic hard material*; *or* b. ≥ 40 cm; and the major part of the interstices between the coarse fragments is filled with *organic material* and the remaining interstices, if present, are void. **HISTOSOLS**	Muusic/ Rockic/ Mawic Cryic Thionic Folic Floatic Subaquatic/ Tidalic Fibric/ Hemic/ Sapric Leptic/ Thyric Murshic/ Drainic Ombric/ Rheic Coarsic Skeletic Andic Vitric	Alcalic/ Dystric/ Eutric Aric Bryic Dolomitic/ Calcaric Fluvic Gelic Hyperorganic Isolatic Lignic Limnic Limonic Mineralic Mulmic Ornithic Placic Pyric Relocatic Salic Sulfidic Technic/ Kalaic Tephric Toxic Transportic Turbic Wapnic

Look only at the RSGs indicated in the Scoresheet!

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Classification according to WRB: third step: decide on the qualifiers

Use the WRB Manual:

https://wrb.isric.org/files/WRB_fourth_edition_2022-12-18_errata_correction_2024-09-24.pdf

5 Definitions of qualifiers

Before using the qualifiers, please read the 'Rules for naming soils' (Chapter 2).

The definitions of the qualifiers for the second-level units relate to RSGs, diagnostic horizons, properties and materials, and to attributes such as colour, chemical conditions, texture, etc. References to the RSGs defined in Chapter 4 and the diagnostics listed in Chapter 3 are shown *in italics*.
Usually, only a limited number of combinations will be possible in a soil name; many of the definitions make the qualifiers mutually exclusive.

General rules

1. **Subqualifiers** (see Chapter 2.3), **which may be used in the soil name instead of the qualifier listed in the Key** (Chapter 4), are found beneath the definition of the respective qualifier (e.g. Protocalcic is found under Calcic). **Subqualifiers, which cannot replace a listed qualifier,** are found in alphabetical order (e.g. Hyperalic).
2. If a subqualifier related to depth requirements can be constructed by the user, **the figure indicates, which rule applies**: (1), (2), (3), (4), (5). If no figure is indicated, these subqualifiers cannot be constructed.

Definitions

Abruptic (ap) (from Latin *abruptus*, broken away): having an *abrupt textural difference* within 100 cm of the mineral soil surface (1).

Geoabruptic (go) (from Greek *gaia*, earth): having an *abrupt textural difference* within 100 cm of the mineral soil surface that is not associated with the upper limit of an *argic*, *natric* or *spodic horizon* (1).

Aceric (ae) (from Latin *acer*, sharp): having within 100 cm of the soil surface a layer with a pH (1:1 in water) between ≥ 3.5 and < 5 and jarosite concentrations (*in Solonchaks only*) (2).

Acric (ac) (from Latin *acer*, sharp): having an *argic horizon* starting ≤ 100 cm from the mineral soil surface with a CEC (by 1 *M* NH_4OAc, pH 7) of < 24 $cmol_c\ kg^{-1}$ clay in some subhorizon within 150 cm of the mineral soil surface; and having exchangeable Al > exchangeable (Ca+Mg+K+Na) in half or more of the depth range between 50 and 100 cm of the mineral soil surface or the lower half of the mineral soil above a limiting layer starting ≤ 100 cm from the mineral soil surface, whichever is shallower (2).
Note: Exchangeable cations are given in $cmol_c\ kg^{-1}$. If these data are not available, pH values may be used according to Annex 2 (Chapter 9.13).

Look only at the qualifiers indicated in the Scoresheet!

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Classification according to WRB: subqualifiers?

In this soil contest, you do not construct subqualifiers.

But we may offer subqualifiers in the scoresheet as a choice.

CEC / kg clay: calculation

CEC: totality of the negative charge that is available per mass to sorb cations:

CEC (soil) = CEC (clay fraction) + CEC (organic fraction)

We divide the CEC by the clay content:
If the content of organic matter is low (as in most subsoil horizons), the error is acceptable.

example:
CEC (given on the pitcard): 12 $cmol_c$ / kg soil [cmol(+) / kg soil]
clay content (your estimation): 30%

calculation:
12/0.3 = 40 result in $cmol_c$ / kg clay

You must not use your own smartphones or calculators.
The organizers will give every team a simple calculator.

CEC / kg clay: soils with argic horizons

CEC ≥ 24 $cmol_c$ / kg clay
(throughout the argic horizon within 150 cm from the mineral soil surface):
RSGs: Alisol, Luvisol
qualifiers: Alic, Luvic

CEC < 24 $cmol_c$ / kg clay
(in some subhorizon of the argic horizon within 150 cm from the mineral soil surface):
RSGs: Acrisol, Lixisol
qualifiers: Acric, Lixic

Do this calculation only for the layers forming an argic horizon!
You don't need to report the figure resulting from your calculation.
If it's obvious, whether the CEC / kg clay is above or below the limit value,
you don't need to do the calculation.

Why are these data important?
low CEC / kg clay: dominance of kaolinite
high CEC / kg clay: dominance of 2:1 clay minerals (and allophanes and imogolites)

mollic / umbric horizons

color:
for the uppermost mineral horizons, the pitcard indicates whether value and chroma, moist, are ≤ 3 or >3

But you have to report in your scoresheet the full color: hue, value, chroma.

dry color: we assume that
value is one chip higher than moist
chroma is the same as moist

base saturation (pH 7):
indicated on the pitcard

Relationships of exchangeable cations (data in $cmol_c$ / kg soil)

sum of base cations ≥ Al
(in the major part from 20 cm from the mineral soil surface to 100 cm or to a limiting layer):
RSGs: Lixisol, Luvisol
qualifiers: Eutric

Al > sum of base cations
(in half or more from 20 cm from the mineral soil surface to 100 cm or to a limiting layer):
RSGs: Acrisol, Alisol
qualifiers: Dystric

All exchangeable cations are given on the pitcard.
If the relationship is not obvious, calculate the sum of base cations.

Coarse fragments (rock fragments): weighted average

Your logo goes here

SCORESHEET Profile ID..... Contestant ID..... Describe.....horizons to a depth of.....cm Nail at.....cm at the bottom of horizon..... (Additional nail at.....cm at the bottom of horizon.....)

1. SITE DESCRIPTION **SCORE:**

Land Use	Landform	Slope		Rock outcrops	Surface rock fragments	Surface unevenness	Litter layer	
Type (1)	*Type (1)*	*Position (1)*	*Gradient % (1)*	*% (1)*	*% (1)*	*Type (1)*	*Area covered % (1)*	*Average thickn. (cm) (1)*

2. PROFILE DESCRIPTION **SCORE:**

Layer boundaries				Texture			Rock fragments		Structure		Color			Carbonates			Redoximorphic features						Aggregate surfaces		Par. mat.
No.	*Lower Depth (cm) (2)*	*Distinct ness (2)*	*Shape (2)*	*Class (2)*	*Clay % (2)*	*Sand % (2)*	*% (2)*	*Weight. Aver. % (2)*	*Type (2)*	*Grade (2)*	*Hue (1)*	*Value (2)*	*Chroma (2)*	*Content (2)*	*Sec. carb. Type (2)*	*Sec. carb. % (1)*	*Oxim. Loc. (2)*	*Reductim. Loc. (2)*	*Oxim. inner % (1)*	*Oxim. outer % (1)*	*Reductim. inner % (1)*	*Reductim. outer % (1)*	*Type (2)*	*% (1)*	*Type (2)*
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weighted average: from the mineral soil surface to 100 cm or to a limiting layer

P. Schad, lecture

Coarse fragments (rock fragments): weighted average: example

Depth (cm)	rock fragments (%)
0 - 20	0
20 - 45	10
45 - 67	25
67 - 105	60

weighted average (0 to 100 cm)
= (20*0 + 25*10 + 22*25 + 33* 60)/100 = 27.8

qualifiers: Skeletic

Anthrosols

defined by the presence of certain diagnostic horizons:
- indicating a significant soil formation caused by land use
- in total ≥ 50 cm thick

Qualifiers: Hydragric/ Irragric/ Hortic/ Plaggic/ Pretic/ Terric

possible in the contest area:
Hydragric Anthrosol (soil resulting from long-term paddy cultivation):
having an *anthraquic horizon* and an underlying *hydragric horizon* with a combined thickness of ≥ 50 cm

Hydragric Anthrosol

example for horizon designation:

Arhp

Arp

Bg

puddled layer

plough pan

anthraquic horizon with gleyic properties, resulting from paddy cultivation

hydragric horizon with stagnic properties, resulting from long-term paddy cultivation

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Gleyic Anthraquic Cambisol

example for horizon designation:

Arhp

Arp

Bwg

Bl

puddled layer

plough pan

anthraquic horizon with gleyic properties, resulting from paddy cultivation

cambic horizon with weakly expressed stagnic properties

natural gleyic properties

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Gleyic properties in the anthraquic horizon

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Redoximorphic features: Location of oximorphic features

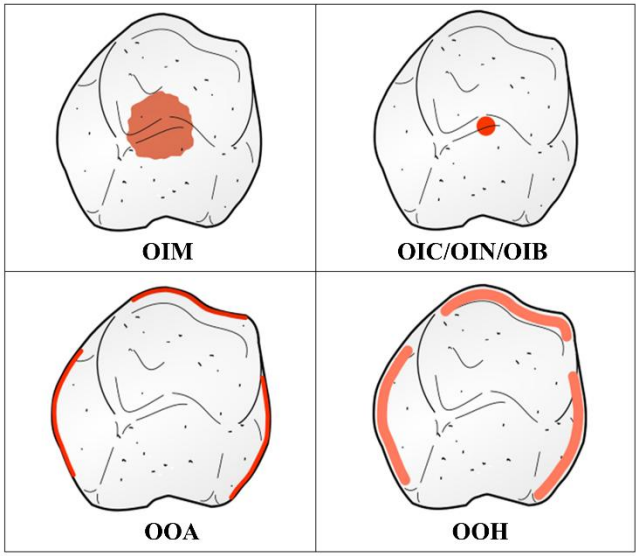
Location		Code
Inner parts	Inside soil aggregates: masses	OIM
	Inside soil aggregates: concretions and/or nodules	OIC
	Inside soil aggregates: combinations of masses, concretions and/or nodules	OIB
	Dominant color, including inner parts of soil aggregates	OID
Outer parts	On surfaces of soil aggregates	OOA
	Adjacent to surfaces of soil aggregates (hypocoats)	OOH
	On biopore walls	OOE
	Dominant color, including outer parts of soil aggregates and/or the biopore walls	OOD
Throughout	Throughout	ORT

I: inner

O: outer

R: random

O: oximorphic



IUSS Working Group WRB, 2022,
Table 8.55, modified,
and Figure 8.19

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Redoximorphic features: Location of reductimorphic features

Location		Code
Inner parts	Inside soil aggregates	RIA
	Dominant color, including inner parts soil aggregates	RID
Outer parts	Outer parts of soil aggregates	ROA
	Around biopores	ROE
	Dominant color, including outer parts of soil aggregates and/or the biopore walls	ROD
Throughout	Throughout	RRT

I: inner

O: outer

R: random

O: reductimorphic

IUSS Working Group WRB, 2022,
Table 8.56, modified

Redoximorphic features: Suggestions: oximorphic features

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1. SITE DESCRIPTION **SCORE:**

Land Use	Landform	Slope		Rock outcrops	Surface rock fragments	Surface unevenness	Litter layer	
Type (1)	*Type (1)*	*Position (1)*	*Gradient % (1)*	*% (1)*	*% (1)*	*Type (1)*	*Area covered % (1)*	*Average thickn. (cm) (1)*

2. PROFILE DESCRIPTION **SCORE:**

Layer boundaries				Texture			Rock fragments		Structure		Color			Carbonates			Redoximorphic features						Aggregate surfaces		Par. mat.
No.	*Lower Depth (cm) (2)*	*Distinctness (2)*	*Shape (2)*	*Class (2)*	*Clay % (2)*	*Sand % (2)*	*% (2)*	*Weight. Aver. % (2)*	*Type (2)*	*Grade (2)*	*Hue (1)*	*Value (2)*	*Chroma (2)*	*Content (2)*	*Sec. carb. Type (2)*	*Sec. carb. % (1)*	*Oxim. Loc. (2)*	*Reductim. Loc. (2)*	*Oxim. inner % (1)*	*Oxim. outer % (1)*	*Reductim. inner % (1)*	*Reductim. outer % (1)*	*Type (2)*	*% (1)*	*Type (2)*
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	Dominant color, including inner parts of soil aggregates	OID
Outer parts	On surfaces of soil aggregates	OOA
	Adjacent to surfaces of soil aggregates (hypocoats)	OOH
	On biopore walls	OOE
	Dominant color, including outer parts of soil aggregates and/or the biopore walls	OOD
Throughout	Throughout	ORT

Task on the scoresheet:

- Indicate the location of the dominant oximorphic feature
- Indicate the % of all inner oximorphic features on the exposed area (profile wall)
- Indicate the % of all outer oximorphic features on the exposed area (profile wall)

(one may be 0%
if the location is ∅, both are 0%)

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Redoximorphic features: Suggestions: reductimorphic features

Your logo goes here

SCORESHEET Profile ID..... Contestant ID..... Describe.....horizons to a depth of.....cm Nail at.....cm at the bottom of horizon..... (Additional nail at.....cm at the bottom of horizon.....)

1. SITE DESCRIPTION **SCORE:**

Land Use	Landform	Slope		Rock outcrops	Surface rock fragments	Surface unevenness	Litter layer	
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2. PROFILE DESCRIPTION **SCORE:**

Layer boundaries				Texture			Rock fragments		Structure		Color			Carbonates			Redoximorphic features						Aggregate surfaces		Par. mat.
No.	*Lower Depth (cm) (2)*	*Distinctness (2)*	*Shape (2)*	*Class (2)*	*Clay % (2)*	*Sand % (2)*	*% (2)*	*Weight. Aver. % (2)*	*Type (2)*	*Grade (2)*	*Hue (1)*	*Value (2)*	*Chroma (2)*	*Content (2)*	*Sec. carb. Type (2)*	*Sec. carb. % (1)*	*Oxim. Loc. (2)*	*Reductim. Loc. (2)*	*Oxim. inner % (1)*	*Oxim. outer % (1)*	*Reductim. inner % (1)*	*Reductim. outer % (1)*	*Type (2)*	*% (1)*	*Type (2)*
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6																									

Location		Code
Inner parts	Inside soil aggregates	RIA
	Dominant color, including inner parts soil aggregates	RID
Outer parts	Outer parts of soil aggregates	ROA
	Around biopores	ROE
	Dominant color, including outer parts of soil aggregates and/or the biopore walls	ROD
Throughout	Throughout	RRT

Task on the scoresheet:

- Indicate the location of the dominant reductimorphic feature
- Indicate the % of all inner reductimorphic features on the exposed area (profile wall)
- Indicate the % of all outer reductimorphic features on the exposed area (profile wall)

(one may be 0%
if the location is ∅, both are 0%)

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Redoximorphic features

Your logo goes here

SCORESHEET Profile ID..... Contestant ID..... Describe.....horizons to a depth of.....cm Nail at.....cm at the bottom of horizon..... (Additional nail at.....cm at the bottom of horizon.....)

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2. PROFILE DESCRIPTION **SCORE:**

Layer boundaries				Texture			Rock fragments		Structure		Color			Carbonates			Redoximorphic features						Aggregate surfaces		Par. mat.
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5																									
6																									

If there is a matrix color left: the % add to < 100%

If there is no matrix color left (all parts of the horizon either oximorphic or reductimorphic): the % add to 100%

The dominant color may be

- a matrix color
- an oximorphic color
- a reductimorphic color

P. Schad, lecture

Videos

https://www.youtube.com/channel/UC-dL7jrqhzwoFfKNLTwGXxQ/videos

P. Schad, lecture